The $RO(C_q)$-Graded Cohomology of Complete Flags in the Regular Representation

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Abstract

Let q be an odd prime and $G = C_q$. We compute the genuine $RO(G)$-graded cohomology of the complete flag variety of the complex regular representation $\mathbb{C}[G]$, with its induced G-action and coefficients in the constant Mackey functor $\underline{\mathbb{F}_q}$. We give an explicit ring presentation with a single Euler-class correction to the elementary symmetric relations and a normalized Schubert basis. At G/G, fixed-point restriction has a nonzero square-zero kernel, while its image is determined by divisibility conditions involving higher powers of the Euler class. We compute the graded module measuring the failure of root-curve edge congruences to determine this image. The proof combines a Schubert module splitting with functorial reconstruction from the even Borel ring and the full coefficient ring, without inverting the Euler class. We also determine the cohomology Mackey functor and the mod-q cohomology ring of the orbit space, whose positive-degree products vanish.

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1 Introduction

Classical Schubert calculus describes torus-equivariant flag cohomology and restriction to torus-fixed points. Passing to the full genuine theory requires control of representation degrees and Euler torsion that Borel comparison can lose. For the cyclic actions considered here, fixed-point restriction has a nonzero kernel, and root-curve edge congruences do not determine its image. We compute both the full genuine ring and this restriction map. The calculation identifies the higher Euler divisibility conditions missing from the edge relations and distinguishes them from the torsion in the restriction kernel.

Let q be an odd prime, let $G = C_q = \langle g \rangle$, and choose the character $\chi(g) = \exp(2\pi i/q)$. Set $V = \mathbb{C}[G]$ and $X = \mathrm{Fl}(V)$. We use the genuine Eilenberg–Mac Lane spectrum $H = H\underline{\mathbb{F}_q}$ of the constant Mackey functor, with full coefficient ring $R = \bigoplus_{\alpha \in RO(G)} R^\alpha$, where $R^\alpha = \pi^G_{-\alpha} \underline{H}$. Write $\lambda = \chi_{\mathbb{R}}$ and $a = e(\lambda) \in R^\lambda$. Throughout, genuine cohomology retains the full $RO(G)$ grading. Unadorned $H^\star_G$ denotes the value at G/G; an underline denotes the cohomology Mackey functor.

The eigenlines $E_j = \chi^{j-1}$, indexed by $1 \le j \le q$, identify the fixed flags with S_q: the flag p_v has ith subspace $E_{v_1} \oplus \cdots \oplus E_{v_i}$. We write $\ell(w)$ for the number of inversions of w, e for the identity permutation, and s_i for the adjacent transposition swapping positions i and $i+1$. Let F_i denote the ith tautological subbundle on X. Put $L_i = F_i/F_{i-1}$ and $z_i = c_1^B(L_i^\vee)$. Let q_X denote the Borel comparison map of Section 4, and let $u \in R^{\lambda-2}$ be the orientation class fixed in Section 2, also denoting its Borel image by u.

Theorem 1.1. *There are unique classes $x_i \in H^\lambda_G(X)$ satisfying $q_X(x_i) = uz_i$. They give an isomorphism*

$$H^\star_G(X) \cong \frac{R[x_1, \dots, x_q]}{(e_1(x), \dots, e_{q-2}(x), e_{q-1}(x) + a^{q-1}, e_q(x))}. \tag{1.1}$$

Here e_j is the jth elementary symmetric polynomial. The ring is free over R on the standard monomials

$$m_{\mathbf{b}} = x_1^{b_1} \cdots x_{q-1}^{b_{q-1}}, \qquad 0 \le b_i \le q - i. \tag{1.2}$$

It is also free on normalized Schubert classes σ_w of degree $\ell(w)\lambda$, $w \in S_q$. For $1 \le i < q$,

$$\sigma_{s_i}\sigma_w = \left(\sum_{j=1}^{i} (j - w_j) \right) a\sigma_w + \sum_{\substack{r \le i < s \\ \ell(wt_{rs}) = \ell(w)+1}} \sigma_{wt_{rs}}, \tag{1.3}$$

where t_{rs} swaps positions r, s and integer coefficients are read in $\mathbb{F}_q$.

Our indexing and dual-line convention give $x_i|_{p_v} = -(v_i - 1)a$. Put $J_d = \mathrm{Ann}_R(a^d)$ for $d \ge 0$, so $J_0 = 0$. Let $i : X^G \hookrightarrow X$ be the fixed-point inclusion, with restriction map

$$i^* : H^\star_G(X) \longrightarrow H^\star_G(X^G) \cong \prod_{v \in S_q} R.$$

Theorem 1.2. *Let M be the normalized Schubert restriction matrix defined in* (6.2) *of Section 6.1, with rows and columns indexed by S_q. In the Schubert basis, i^* has matrix*

$$T = M \,\mathrm{diag}(a^{\ell(w)}), \qquad M \in GL_{q!}(\mathbb{F}_q). \tag{1.4}$$

Consequently

$$\ker i^* = \bigoplus_{w \ne e} J_{\ell(w)}\sigma_w, \qquad f \in \mathrm{im}\, i^* \iff (M^{-1}f)_w \in a^{\ell(w)}R \text{ for every } w. \tag{1.5}$$

The kernel is nonzero and has square zero at G/G. Restriction becomes an isomorphism after inverting a. If $\Gamma \subset R^{q!}$ is the module of tuples satisfying the root-curve edge congruences, then

$$\Gamma/\operatorname{im} i^* \cong \bigoplus_{\ell(w)\geq 2} \frac{aR}{a^{\ell(w)}R} \tag{1.6}$$

as graded R-modules, with the inherited ideal gradings on the right.

The presentation (1.1) holds over the full coefficient ring R, without inverting a. The ordinary Chern classes are denoted by $y_i = c_1(L_i^\vee)$. Defining the genuine classes x_i by comparison does not assume that the equivariant line bundles L_i admit Thom classes in one fixed representation degree.

The kernel in (1.5) records Euler torsion in R. The edge congruences defining Γ require $f_v - f_{vt_{rs}} \in aR$, whereas the image criterion imposes the higher powers $a^{\ell(w)}$ after applying M^{-1}. These higher divisibility obstructions already occur for tuples with polynomial coefficients in a, as the example in Remark 6.1 shows. They are therefore distinct from the negative-cone torsion responsible for the restriction kernel. The quotient in (1.6) measures precisely what the edge congruences omit, for every odd prime q. The same calculation determines the mod-q cohomology of X/G: Corollary 7.1 gives its Betti numbers, and every product of positive-degree classes is zero.

Proof strategy

The Schubert filtration has representation cells with nontrivial complex characters. After smashing with H, faithful character suspensions become equivalent to λ-suspensions. For $q > 3$, these are H-module equivalences, not identities in $RO(G)$. The connecting components lie in $R^{1-d\lambda} = 0$ for $d \geq 1$, so Proposition 3.1 gives

$$H \wedge X_+ \simeq \bigvee_{w\in S_q} \Sigma^{\ell(w)\lambda} H.$$

This splitting controls the representation degrees; it is neither chosen naturally nor required to preserve multiplication.

The reconstruction step is given by Propositions 4.1 and 4.2. For a space Y of finite G-CW type admitting a finite H-module splitting with shifts in nonnegative multiples of λ, Borel comparison is an isomorphism in every pure $n\lambda$ degree. These isomorphisms respect multiplication and equivariant pullbacks, and recover the full genuine ring from the even Borel subring together with R. Borel degree $2n$ is regraded as $n\lambda$, with $t = c_1(\chi) \in H^2(BG;\mathbb{F}_q)$ sent to a. Section 3 verifies the finite G-CW hypothesis for X separately from the representation-cell filtration. Classical flag-bundle and Schubert formulas then give the ring and its normalized basis. Since the finite set X^G satisfies the same splitting hypothesis, naturality for $i : X^G \hookrightarrow X$ identifies the restriction map itself, including its Euler-torsion kernel, and proves Theorem 1.2.

Related work

Basu and Ghosh compute the constant-$\mathbb{F}_q$ coefficients and prove freeness for sparse complexes formed from representation cells [2, §3.2 and Theorem 4.12]. Basu, Dey, and Karmakar compare faithful character suspensions in [4, Propositions 2.10–2.11] and [3, Proposition 3.6]; they also establish module splittings for projectivizations of multiples of the regular representation [4, Theorem 4.7]. We use their coefficient and suspension results, while proving the splitting for flags directly. The remaining Borel calculations use classical flag-bundle theory and Schubert calculus: the divided-difference construction of Bernstein, Gel'fand, and Gel'fand [5], and the flag-bundle and double Schubert formulas presented by Anderson and Fulton [1, Chapters 4 and 10].

The torus-equivariant representation-cell construction recalled by Harada, Henriques, and Holm supplies the geometric filtration [9, §5]. The Euler-class hypotheses of their GKM image theorem [9, Theorem 3.1] fail here: the root Euler classes are zero divisors and generate the same ideal aR. The coefficient calculations and classical Schubert formulas provide the input for our computation. The reconstruction above realizes these formulas over the full genuine coefficient ring and identifies fixed-point restriction before Euler localization. Its kernel, its higher Euler divisibility conditions, and the explicit defect (1.6) give a uniform description of the information beyond root-curve edge congruences for this family of flag varieties.

Section 2 records the coefficient calculations, Section 3 proves the Schubert module splitting, and Section 4 establishes comparison and reconstruction. Sections 5 and 6 give the ring, Schubert multiplication, and fixed-point restriction. Section 7 determines the Mackey structure, works out the case $q = 3$, and derives the mod-q cohomology of X/G. Appendix A gives the coefficient transfer normalization.

2 Conventions and coefficient calculations

We use genuine G-spectra indexed on a complete G-universe. Representation suspensions, function spectra, and duality are understood in this stable category; see [12, Chapters I–III].

2.1 Representation degrees and orientations

Put $m = (q-1)/2$ and

$$\lambda_r = (\chi^r)_{\mathbb{R}} \quad (1 \le r \le m), \qquad \lambda = \lambda_1, \qquad RO(G) = \mathbb{Z}\{1, \lambda_1, \dots, \lambda_m\}.$$

The representation degree of a class is cohomological. Its parity is the parity of its underlying real dimension. For $1 \le r \le m$ the map $z \mapsto z^r$ extends to $S^\lambda \to S^{\lambda_r}$. On underlying spheres it has degree r, and on fixed spheres S^0 it is the identity. After smashing with H it is an equivalence: the underlying map is an equivalence modulo q, and applying geometric fixed points gives the identity on $\Phi^G H$. These two tests detect equivalences for C_q. Equivalently, faithful character suspensions can be compared by [3, Proposition 3.6], followed by derived extension of scalars from $H\underline{\mathbb{Z}}$ to $H\underline{\mathbb{F}_q}$.

Let $u_r \in R^{\lambda_r - 2}$ be the mod-q complex orientation class of λ_r [2, Definition 3.3], and put $u = u_1$. Choose units $\varepsilon_r \in R^{\lambda_r - \lambda}$ with $\varepsilon_1 = 1$. Normalize the restrictions of u_r and ε_r to the scalar 1 under the identifications of their respective underlying coefficient groups with $\mathbb{F}_q$ supplied by the complex orientations. These scalar identifications retain the $RO(G)$-degree at G/e; they do not identify elements of different degrees with the ring unit. The class u is not a unit in R. With these compatible choices,

$$H \wedge S^{\lambda_r} \simeq \Sigma^\lambda H, \qquad u_r = \varepsilon_r u, \qquad a_{\lambda_r} = r\varepsilon_r a. \tag{2.1}$$

The unit ε_r is represented by the H-module map induced by $z \mapsto z^r$, divided by $r \in {\mathbb{F}_q}^\times$. These conventions agree with [4, Propositions 2.10–2.11] after reduction modulo q. For $q = 3$ there are no additional units. In general, if R_λ denotes the $\mathbb{Z}\{1, \lambda\}$ subgrading, then

$$R = R_\lambda[\varepsilon_2^{\pm 1}, \dots, \varepsilon_m^{\pm 1}]. \tag{2.2}$$

Indeed every degree has a unique expression in $1, \lambda, \lambda_2 - \lambda, \dots, \lambda_m - \lambda$; multiplication by the corresponding unit identifies its group with one in the two-direction subgrading.

For a character χ^b with $m < b \le q-1$, complex conjugation identifies its real representation with λ_{q-b} and reverses complex orientation. We will use the normalized pure-λ Euler factor ba, determined by Borel comparison. This convention distinguishes normalized factors from the transport of canonical Euler classes under a real representation isomorphism.

2.2 Coefficients in the two-direction subgrading

The coefficient calculation of [2, §3.2, pp. 39–41, especially Proposition 3.4], in cohomological degrees, is

$$R_\lambda = P \oplus N_\lambda, \qquad P = \mathbb{F}_q[a,u] \otimes \Lambda(\kappa), \qquad |u| = \lambda - 2, \quad |\kappa| = \lambda - 1. \tag{2.3}$$

The negative cone has basis $\nu_{j,k}, \mu_{j,k}$ for $j, k \geq 1$, with

$$|\nu_{j,k}| = 1 + 2j - (j+k)\lambda, \qquad |\mu_{j,k}| = 2j + (1-j-k)\lambda. \tag{2.4}$$

These generators correspond to the two families in the Tate cokernel computed immediately before that proposition. We choose their signs so that $\kappa\nu_{j,k} = \mu_{j,k}$. The module structure of this cokernel gives

$$\begin{gathered} a\nu_{j,k} = \nu_{j,k-1}, \quad u\nu_{j,k} = \nu_{j-1,k}, \quad a\mu_{j,k} = \mu_{j,k-1}, \quad u\mu_{j,k} = \mu_{j-1,k}, \\ \kappa\nu_{j,k} = \mu_{j,k}, \qquad \kappa\mu_{j,k} = 0. \end{gathered} \tag{2.5}$$

A symbol with a zero index means zero. The symbols are classes, not fractions in R. Let $\mathcal{L} = \mathbb{F}_q[\varepsilon_2^{\pm 1}, \dots, \varepsilon_m^{\pm 1}]$ and $N = \mathcal{L} \otimes_{\mathbb{F}_q} N_\lambda$.

Lemma 2.1. *The coefficient groups used below satisfy*

$$R^{1-d\lambda} = 0 \quad (d \geq 0), \qquad R^{d\lambda} = \begin{cases} \mathbb{F}_q a^d, & d \geq 0, \\ 0, & d < 0. \end{cases} \tag{2.6}$$

For $d \geq 1$,

$$J_d = \mathcal{L} \otimes_{\mathbb{F}_q} \operatorname{span}_{\mathbb{F}_q}\{\nu_{j,k}, \mu_{j,k} : j \geq 1,\ 1 \leq k \leq d\}. \tag{2.7}$$

The ideal N has square zero, and $aN = N$.

Proof. In the subgrading, a positive monomial $a^i u^j \kappa^\eta$ has degree $(-2j - \eta) + (i + j + \eta)\lambda$. Its trivial-representation coefficient is never 1 and is 0 only if $j = \eta = 0$. The corresponding coefficient of every negative-cone generator is at least 2. Additional representation directions cannot occur in a pure-λ degree. This proves (2.6). Multiplication by a is injective on $\mathcal{L} \otimes P$; on N it shifts the second index. This gives (2.7) and $aN = N$. If $a^d n = 0$ and $n' = a^d z$ with $z \in N$, then $nn' = 0$. Every element of N is killed by a power of a, so $N^2 = 0$. □

Choose $s \in H^1(BG; \mathbb{F}_q)$ so that $t = \beta s = c_1(\chi)$. The Borel coefficient ring in the full grading is

$$R_B = \mathbb{F}_q[t] \otimes \Lambda(s)[u^{\pm 1}, \varepsilon_2^{\pm 1}, \dots, \varepsilon_m^{\pm 1}], \qquad |s| = 1, \quad |t| = 2.$$

We use the same letters for images of u and ε_r. The coefficient comparison is

$$q_R(a) = ut, \quad q_R(u) = u, \quad q_R(\kappa) = us, \quad q_R(\varepsilon_r) = \varepsilon_r, \quad q_R(N) = 0. \tag{2.8}$$

The two-direction comparison is computed in [2, §3.2]; adjoining the units ε_r gives (2.8). The negative cone maps to zero because it is u-torsion and u is invertible in R_B. Thus $R_B = R[u^{-1}]$, but u is not a unit in R. In particular,

$$q_R : R^{d\lambda} \xrightarrow{\cong} R_B^{d\lambda} \quad (d \in \mathbb{Z}), \qquad R_B^{d\lambda} = \begin{cases} \mathbb{F}_q (ut)^d, & d \geq 0, \\ 0, & d < 0. \end{cases} \tag{2.9}$$

To check the Borel group in this formula, a monomial $u^k t^h s^\eta$ has degree $(2h + \eta - 2k) + k\lambda$, with $h \geq 0$ and $\eta \in \{0, 1\}$. Degree $d\lambda$ forces $k = d$, $\eta = 0$, and $h = d \geq 0$. The exponents of all additional representation units must be zero. Thus the only possible monomial is $(ut)^d$, which is the nonzero image of a^d. No Euler localization is used in this comparison.

3 Schubert cells and the H-module splitting

Write $V = \bigoplus_{j=1}^{q} E_j$, where $E_j \cong \chi^{j-1}$, and choose eigenvectors e_j. We compose permutations as functions: $(vw)(i) = v(w(i))$. Thus right multiplication by t_{rs} swaps positions. Let $\ell(w)$ count inversions and let w_0 be the longest permutation. Every invariant subspace is a sum of eigenlines, so

$$X^G = \{p_w : w \in S_q\}, \qquad (p_w)_i = E_{w_1} \oplus \cdots \oplus E_{w_i}. \tag{3.1}$$

Let $\mathcal{B}$ be the upper triangular Borel subgroup of $GL(V)$. The echelon coordinates on $\mathcal{B}p_w$ are

$$v_i = e_{w_i} + \sum_{\substack{j>i \\ w_j<w_i}} z_{ji} e_{w_j}, \qquad F_i = \langle v_1, \dots, v_i \rangle.$$

Applying g and normalizing the pivots multiplies z_{ji} by $\chi^{w_j - w_i}(g)$. Hence this cell is the complex representation

$$W_w = \bigoplus_{\substack{i<j \\ w_i>w_j}} \chi^{w_j - w_i}. \tag{3.2}$$

All its characters are nontrivial, and $\dim_{\mathbb{R}} W_w = 2\ell(w)$.

The Bruhat closure relation gives the closed filtration

$$X_{\le d} = \bigcup_{\ell(w)\le d} \mathcal{B}p_w, \qquad 0 \le d \le q(q-1)/2.$$

The closure order is standard [7, Proposition 1.2.1(iii)]. The torus-equivariant Schubert attachment construction [9, §§2 and 5], restricted to G, makes these inclusions equivariant cofibrations: the boundary inclusion of each representation disk is a G-cofibration, and pushouts and finite composites preserve this property. Collapsing the preceding closed stage compactifies each cell of length d separately. Indeed the boundary of each such cell in its Schubert closure lies in $X_{\le d-1}$, and the closures of distinct length-d cells intersect only in this lower stage. Since there are finitely many cells and the stage is compact, the quotient is the wedge of their one-point compactifications:

$$X_{\le d}/X_{\le d-1} \cong \bigvee_{\ell(w)=d} S^{W_w} \qquad (d \ge 1). \tag{3.3}$$

At $d = 0$ the based stage is S^0.

Proposition 3.1. *There is an equivalence of genuine H-modules*

$$H \wedge X_+ \simeq \bigvee_{w\in S_q} \Sigma^{\ell(w)\lambda} H. \tag{3.4}$$

Proof. Put $U_d = H \wedge (X_{\le d})_+$ and $Q_d = \bigvee_{\ell(w)=d} \Sigma^{d\lambda} H$. Smashing (3.3) with H and applying (2.1) gives a triangle of H-modules

$$U_{d-1} \longrightarrow U_d \longrightarrow Q_d \xrightarrow{\delta_d} \Sigma U_{d-1}.$$

Suppose inductively that U_{d-1} is a wedge of $\Sigma^{j\lambda} H$ with $j < d$. Since both wedges are finite, δ_d is determined by its matrix of components. Each lies in

$$[\Sigma^{d\lambda} H, \Sigma^{j\lambda+1} H]^G_{H\text{-mod}} = R^{1-(d-j)\lambda} = 0.$$

Thus δ_d is null. Exactness after applying $[Q_d, -]^G_{H\text{-mod}}$ lifts the identity of Q_d to a section $Q_d \to U_d$. Together with $U_{d-1} \to U_d$, this gives an equivalence $U_{d-1} \vee Q_d \simeq U_d$. Starting from $U_0 = H$ proves the assertion with the indicated representation shifts. □

This is a module splitting. It supplies the degrees of an additive basis; its choice is neither natural nor required to preserve products.

The filtration above uses representation cells. Ordinary G-CW cells have the form $G/K \times D^n$, with trivial action on the disk. The finite G-CW hypothesis of Proposition 4.1 follows separately: X is a compact smooth manifold with a smooth G-action, so Illman's equivariant triangulation theorem and its CW consequence [11, Theorem 7.1 and Corollary 7.2] give a G-CW structure on X. Its quotient CW complex X/G is compact, hence has finitely many cells. Thus this G-CW structure is finite.

4 Comparison with Borel cohomology and reconstruction

For a finite G-CW complex Y, comparison is induced by

$$q_Y : F(Y_+, H) \longrightarrow F(EG_+, F(Y_+, H)).$$

Write $H^*_B(Y) = H^*(EG \times_G Y; \mathbb{F}_q)$ in ordinary integer degrees. The orientations identify the full representation-graded Borel group in degree $n\lambda$ with $u^n H^{2n}_B(Y)$.

Proposition 4.1. *Suppose Y has finite G-CW type and*

$$H \wedge Y_+ \simeq \bigvee_{j=1}^{h} \Sigma^{d_j\lambda} H, \qquad d_j \geq 0. \tag{4.1}$$

Then the comparison map gives isomorphisms

$$q_Y : H^{n\lambda}_G(Y) \xrightarrow{\cong} u^n H^{2n}_B(Y) \qquad (n \in \mathbb{Z}). \tag{4.2}$$

These isomorphisms commute with products and equivariant pullbacks.

Proof. The free-module adjunction and the finite splitting identify

$$F(Y_+, H) \simeq F_H(H \wedge Y_+, H) \simeq \bigvee_j \Sigma^{-d_j\lambda} H.$$

Here F_H is the internal function spectrum of H-modules. The function spectrum out of a finite wedge is a finite product, hence a wedge, and $F_H(\Sigma^{d_j\lambda}H, H) \simeq \Sigma^{-d_j\lambda}H$. Naturality of the map $C \to F(EG_+, C)$ with respect to this H-module equivalence identifies comparison in degree $n\lambda$ with the direct sum of coefficient maps $R^{(n-d_j)\lambda} \to R_B^{(n-d_j)\lambda}$. Each is an isomorphism by (2.9). Multiplication and pullback commute with the original comparison map, so the claimed compatibilities follow without a multiplicative choice of splitting. □

Let $\mathcal{H}_\lambda(Y) = \bigoplus_{n\geq 0} H^{n\lambda}_G(Y)$. For an even Borel class $b \in H^{2n}_B(Y)$, let $\mathcal{D}b$ have degree $n\lambda$; $\mathcal{D}H^{\mathrm{even}}_B(Y)$ is the even Borel subring with this grading.

Proposition 4.2. *Under* (4.1)*, multiplication and comparison give natural graded ring isomorphisms*

$$H^\star_G(Y) \cong R \otimes_{\mathbb{F}_q[a]} \mathcal{H}_\lambda(Y) \cong R \otimes_{\mathbb{F}_q[t]} \mathcal{D}H^{\mathrm{even}}_B(Y), \qquad t \longmapsto a. \tag{4.3}$$

The last isomorphism sends $r \otimes \mathcal{D}b$ to $rq_Y^{-1}(u^n b)$ when $b \in H^{2n}_B(Y)$.

Proof. Dualizing the splitting gives an R-basis β_j with $|\beta_j| = d_j\lambda$. Equation (2.6) gives, on this same basis,

$$\mathcal{H}_\lambda(Y) = \bigoplus_j \mathbb{F}_q[a]\beta_j, \qquad H^\star_G(Y) = \bigoplus_j R\beta_j.$$

Consequently multiplication sends $r \otimes a^k\beta_j$ to $ra^k\beta_j$ and is an isomorphism on each free summand. It is a ring map because the tensor product is taken along the coefficient inclusion $\mathbb{F}_q[a] \to \mathcal{H}_\lambda(Y)$, and the coefficients are central with the usual graded sign. Proposition 4.1 identifies the pure subring with the regraded Borel subring: for $b \in H^{2n}_B(Y)$, put $\theta_Y(b) = q_Y^{-1}(u^n b)$. Comparison of products gives $\theta_Y(bc) = \theta_Y(b)\theta_Y(c)$, while $\theta_Y(tb) = a\theta_Y(b)$ by $q_R(a) = ut$. Thus $r \otimes \mathcal{D}b \mapsto r\theta_Y(b)$ is precisely the second isomorphism.

For an equivariant map $f : Z \to Y$ between spaces satisfying (4.1), naturality of comparison gives

$$q_Z(f_G^*\theta_Y(b)) = u^n f_B^* b = q_Z(\theta_Z(f_B^* b)).$$

Injectivity in degree $n\lambda$ proves equality of these lifts. Extending by the coefficients proves naturality on the full rings. □

No flatness of R over $\mathbb{F}_q[a]$ is needed. Both X and the finite set X^G satisfy (4.1); hence the reconstruction commutes with their inclusion. This will identify fixed-point restriction itself, including its Euler torsion.

5 The ring presentation and equivariant Schubert calculus

5.1 The Borel flag bundle

The Borel construction is the complete flag bundle of $\mathcal{V} = EG \times_G V$ over BG. Put $z_i = c_1^B(L_i^\vee)$. The roots of $\mathcal{V}^\vee$ are $r_j t$, with $r_j = -(j-1) \in \mathbb{F}_q$. Since these range over $\mathbb{F}_q$,

$$\prod_{j=1}^{q}(T - r_j t) = T^q - t^{q-1}T. \tag{5.1}$$

The flag-bundle formula gives

$$H^*_B(X) = \frac{\mathbb{F}_q[t, z_1, \dots, z_q] \otimes \Lambda(s)}{(e_1(z), \dots, e_{q-2}(z), e_{q-1}(z) + t^{q-1}, e_q(z))}. \tag{5.2}$$

For completeness, this formula follows by successively choosing a line in the remaining quotient bundle. The ordinary projective-bundle formula and Leray–Hirsch give the basis $z_1^{b_1} \cdots z_{q-1}^{b_{q-1}}$, $0 \le b_i \le q - i$, over $H^*(BG; \mathbb{F}_q)$; see [1, Proposition 4.4.1] and [10, Theorem 4D.1]. The Chern identities give the displayed relations. The monic-extension argument below proves that their quotient has precisely this basis, so the map is an isomorphism. This uses no division by $q!$. Every basis element has even degree; thus the even subring of (5.2) is its part without s.

5.2 Genuine generators and the monomial basis

Define x_i by the unique lift

$$q_X(x_i) = uz_i. \tag{5.3}$$

The relations in (1.1) compare to u^j times their Borel counterparts and vanish by (5.2). They lie in pure degrees where comparison is injective. Proposition 4.2 identifies their quotient with the complete ring.

We prove the basis assertion over a graded-commutative coefficient ring K. The polynomial variables are central and have even parity; the ordered factorizations below have roots of this parity. Start with a monic polynomial $f_0(T)$ of degree q with even coefficients over $K_0 = K$. Recursively set

$$K_i = K_{i-1}[x_i]/(f_{i-1}(x_i)), \qquad f_{i-1}(T) = (T - x_i)f_i(T) \quad \text{in } K_i[T].$$

The second equality is monic division: its remainder is $f_{i-1}(x_i) = 0$. At stage i the monic relation has degree $q - i + 1$, so $1, x_i, \dots, x_i^{q-i}$ form a K_{i-1}-basis. Iterating gives the basis (1.2) of K_q; the last root is determined by its linear relation. An ordered factorization of f_0 in a graded-commutative K-algebra with even roots factors uniquely through these stages. Thus K_q has the same universal property as the quotient imposing the elementary-symmetric coefficients of f_0, and the two presentations are isomorphic. Apply this construction to $f_0(T) = T^q - a^{q-1}T$ over R. The signs of the coefficients give exactly (1.1). This proves the first assertion of Theorem 1.1, including the basis. Monic division remains valid when R has zero divisors.

5.3 Schubert conventions and divisor products

We use opposite Schubert classes relative to the upper triangular $\mathcal{B}$. To specify signs, let $\mathfrak{S}_w(Z; B)$ be the double Schubert polynomials with

$$\mathfrak{S}_{w_0}(Z; B) = \prod_{i+j \le q} (Z_i - B_j), \qquad \partial_i f = \frac{f - s_i f}{Z_i - Z_{i+1}}, \qquad \partial_i \mathfrak{S}_w = \mathfrak{S}_{ws_i} \quad \text{if } \ell(ws_i) < \ell(w). \tag{5.4}$$

Here s_i acts on the Z variables, and the quotient is a polynomial. The divided-difference operators satisfy $\partial_i^2 = 0$ and the Coxeter braid relations [5, Theorem 3.4]. The numerator $f - s_i f$ vanishes when $Z_i = Z_{i+1}$, so it is divisible by $Z_i - Z_{i+1}$ over the integral polynomial ring. These operators and their relations therefore remain defined after reduction modulo q. The descending construction is independent of the chosen reduced word. The classical Schubert-basis theorem identifies these polynomials with the opposite classes in the torus-equivariant flag ring [1, §10.10.1]. Here "opposite" is relative to the upper-triangular group $\mathcal{B}$: these are the lower-triangular-invariant classes in that reference. For the diagonal torus T, its variables are $Z_i = -c_1^T(L_i)$ and $B_j = -c_1^T(E_j)$, in agreement with our conventions. Specialize $B_j = r_j t$ and put

$$S_w = \mathfrak{S}_w(z; r_1 t, \dots, r_q t), \qquad q_X(\sigma_w) = u^{\ell(w)} S_w. \tag{5.5}$$

Under the fiber inclusion $X \hookrightarrow EG \times_G X$, the classes S_w restrict to $\mathfrak{S}_w(y; 0)$, where $y_i = c_1(L_i^\vee)$. These are the ordinary Schubert classes and form an $\mathbb{F}_q$-basis of $H^*(X; \mathbb{F}_q)$: the integral Schubert basis comes from the even-dimensional Schubert cell decomposition, so reduction modulo q preserves it. Leray–Hirsch [10, Theorem 4D.1] therefore makes the S_w an $H^*(BG; \mathbb{F}_q)$-basis of $H_B^*(X)$. Each S_w has even degree, and $H^*(BG; \mathbb{F}_q) = \mathbb{F}_q[t] \otimes \Lambda(s)$, so they also form an $\mathbb{F}_q[t]$-basis of $H_B^{\mathrm{even}}(X)$. Reconstruction now shows that the σ_w form an R-basis. Homogeneity also gives the polynomial formula

$$\sigma_w = \mathfrak{S}_w(x; r_1 a, \dots, r_q a).$$

For a simple reflection,

$$\sigma_{s_i} = x_1 + \cdots + x_i + \binom{i}{2} a. \tag{5.6}$$

In the classical equivariant Monk formula, the coefficient of S_w is $\sum_{j=1}^{i}(B_{w_j} - B_j)$, while each Bruhat cover crossing position i has coefficient 1; see [1, Theorem 10.7.3, p. 170]. After $B_j = -(j-1)t$, the first coefficient is $\sum_{j=1}^{i}(j - w_j)t$. Multiplying by $u^{\ell(w)+1}$ and using injectivity of comparison in this degree proves (1.3). This completes Theorem 1.1.

6 Fixed-point restriction and edge congruences

6.1 The restriction matrix

At p_v, the dual line $L_i^\vee$ has character $\chi^{-(v_i-1)}$. Naturality of (5.3) therefore gives

$$z_i|_{p_v} = r_{v_i} t, \qquad x_i|_{p_v} = r_{v_i} a. \tag{6.1}$$

In the Schubert basis define

$$M_{v,w} = \mathfrak{S}_w(r_{v_1}, \dots, r_{v_q}; r_1, \dots, r_q) \in \mathbb{F}_q. \tag{6.2}$$

Homogeneity gives $\sigma_w|_{p_v} = a^{\ell(w)} M_{v,w}$. Classical Schubert restriction has Bruhat support and diagonal

$$M_{v,w} = 0 \quad \text{unless } w \le v, \qquad M_{w,w} = \prod_{\substack{i<j \\ w_i > w_j}} (w_j - w_i). \tag{6.3}$$

These are the normal Euler factors of the opposite Schubert class at its own fixed point; they also follow from the polynomial subword restriction formula [6, Theorem 4 and §8, Remark 1]. In that reference the Schubert normalization is $\overline{K}_w = \pi_w(O) K_w$. The resulting root-product identities hold over the integral polynomial ring; we specialize $B_j = -(j-1)$ over $\mathbb{Z}$ and then reduce modulo q. Every diagonal factor in (6.3) is nonzero modulo q. Ordering permutations by a linear extension of Bruhat order makes M triangular with invertible diagonal. Thus $M \in GL_{q!}(\mathbb{F}_q)$ and (1.4) follows.

One can alternatively use the standard monomials. If $d_{\mathbf{b}} = \sum_i b_i$, their matrix is

$$\mathsf{E}\,\mathrm{diag}(a^{d_{\mathbf{b}}}), \qquad \mathsf{E}_{v,\mathbf{b}} = \prod_{i=1}^{q-1} r_{v_i}^{b_i}. \tag{6.4}$$

To see that E is invertible, introduce an independent parameter τ in the ordered splitting algebra of $T^q - \tau^{q-1} T$ over $\mathbb{F}_q[\tau]$ and set $\tau = 1$. The polynomial $T^q - T$ has distinct roots. Successive applications of the Chinese remainder theorem identify its ordered splitting algebra with $\prod_{v \in S_q} \mathbb{F}_q$. Its monomial basis has evaluation matrix E, proving the claim.

6.2 Kernel and actual image

For $h = \sum_w c_w \sigma_w$, equation (1.4) gives

$$(M^{-1} i^* h)_w = a^{\ell(w)} c_w.$$

This proves the kernel and image in (1.5); it also gives

$$\operatorname{coker} i^* \cong \bigoplus_{w \neq e} R/a^{\ell(w)} R \tag{6.5}$$

with the target grading. All the source kernel coefficients lie in N, so Lemma 2.1 gives $(\ker i^*)^2 = 0$. Since $\mu_{1,1} \neq 0$ and $a\mu_{1,1} = 0$, the basis gives the nonzero kernel element $\mu_{1,1}\sigma_{s_1}$. After inverting a, the diagonal matrix in (1.4) is invertible; hence so is restriction. This proves the corresponding assertions of Theorem 1.2.

6.3 Root-curve conditions and their defect

For $r < s$, the standard torus root curve joining p_v and $p_{vt_{rs}}$ is G-equivariantly $\mathbb{P}(E_{v_r} \oplus E_{v_s})$. The affine coordinate z defined by the line $\langle e_{v_r} + z e_{v_s} \rangle$ has character $\chi^{v_s - v_r}$. Thus the curve is equivariantly S^ρ, with $\rho = (\chi^{v_s - v_r})_{\mathbb{R}}$, $p_0 = p_v$, and $p_\infty = p_{vt_{rs}}$. The exponent is read modulo q, with the complex orientation and Euler normalization of Section 2. Using the infinity pole p_∞ as basepoint, the unreduced cohomology splits as

$$H^\alpha_G(S^\rho) \cong R^\alpha \oplus \widetilde{H}^\alpha_G(S^\rho) \cong R^\alpha \oplus R^{\alpha - \rho}.$$

With the poles ordered as (p_∞, p_0), restriction sends (c, d) to $(c, c + a_\rho d)$, since the zero pole pulls back reduced cohomology by the Euler map $S^0 \to S^\rho$. Thus their difference is divisible by a_ρ, even when that Euler class is a zero divisor. After multiplication by the relevant unit in (2.1), the Euler ideal is aR. Thus define

$$\Gamma = \{ f \in R^{q!} : f_v - f_{vt_{rs}} \in aR \text{ for every } v \text{ and } r < s \}.$$

The transposition graph is connected, so

$$\Gamma = R\mathbf{1} + aR^{q!}. \tag{6.6}$$

The identity Schubert column of M is $\mathbf{1}$. As a constant automorphism of the target $R^{q!}$, M^{-1} preserves its grading. It therefore gives

$$M^{-1}\Gamma = R \oplus \bigoplus_{w \neq e} aR, \qquad M^{-1} \operatorname{im} i^* = \bigoplus_w a^{\ell(w)} R.$$

Taking the quotient proves (1.6). The coordinate change acts on the target, whose summands all have the same grading; no shift by $\ell(w)\lambda$ is added to these ideal quotients. This is a module isomorphism, not a coordinatewise ring isomorphism.

The defect is nonzero already in degree λ: if $\ell(w) \geq 2$, $a \notin a^{\ell(w)}R$ by (2.6). Moreover $aN = N$ implies that the negative cone vanishes in each defect summand. Euler torsion causes the restriction kernel, whereas first-order edge congruences fail to impose the higher Euler powers in the image. The edge obstruction already occurs for tuples in $\mathbb{F}_q[a]^{q!}$.

Remark 6.1. Put $d_{\max} = \ell(w_0) = q(q-1)/2$ and define $f \in R^{q!}$ by $f_{w_0} = a$ and $f_v = 0$ for $v \neq w_0$. Every edge difference is 0 or $\pm a$, so $f \in \Gamma$. By Bruhat support, the w_0 column of M is supported only at w_0. Writing $m_0 = M_{w_0, w_0} \in {\mathbb{F}_q}^\times$, we obtain $(M^{-1}f)_{w_0} = m_0^{-1}a$ and $(M^{-1}f)_w = 0$ for $w \neq w_0$. The image criterion (1.5) would require $m_0^{-1}a \in a^{d_{\max}}R$. This is impossible: the degree-λ part of that ideal is $a^{d_{\max}}R^{(1-d_{\max})\lambda} = 0$ by (2.6), whereas $m_0^{-1}a \neq 0$. Thus f satisfies all edge relations but does not belong to $\operatorname{im} i^*$ for any odd prime q.

For an equivariantly formal complex projective variety with an algebraic torus action, finitely many torus fixed points, and finitely many one-dimensional torus orbits, the classical GKM theorem describes ordinary equivariant cohomology with real coefficients by edge conditions [8, Theorem 1.2.2]. The generalized image theorem in [9, Theorem 3.1] requires non-zero-divisor Euler classes and suitable relative-primality conditions. Here $a\mu_{1,1} = 0$, and all nontrivial character Euler ideals are aR. Thus the explicit image above contains divisibility conditions that the edge congruences alone do not impose.

7 Mackey structure and specializations

7.1 Restriction and transfer

To record the full grading at G/e, put $v = \mathrm{res}(u)$ and $\bar{\varepsilon}_r = \mathrm{res}(\varepsilon_r)$. These symbols have underlying dimension zero, but retain their representation degrees. The coefficient ring is

$$\underline{R}^\star(G/e) = \mathbb{F}_q[v^{\pm 1}, \bar{\varepsilon}_2^{\pm 1}, \dots, \bar{\varepsilon}_m^{\pm 1}], \qquad |v| = \lambda - 2, \quad |\bar{\varepsilon}_r| = \lambda_r - \lambda.$$

Let $\bar{\varepsilon}^{\mathbf{n}} = \prod_{r=2}^m \bar{\varepsilon}_r^{n_r}$. Restriction on coefficients sends a, κ, N to zero and sends u, ε_r to $v, \bar{\varepsilon}_r$. With the normalization of Appendix A,

$$\mathrm{tr}_R(v^n \bar{\varepsilon}^{\mathbf{n}}) = \begin{cases} 0, & n \geq 0, \\ \varepsilon^{\mathbf{n}} \mu_{-n,1}, & n < 0. \end{cases} \tag{7.1}$$

For $y_i = c_1(L_i^\vee)$ in ordinary degree 2, the lower-orbit ring is

$$\underline{H}^\star_G(X)(G/e) = \frac{\mathbb{F}_q[v^{\pm 1}, \bar{\varepsilon}_2^{\pm 1}, \dots, \bar{\varepsilon}_m^{\pm 1}, y_1, \dots, y_q]}{(e_1(y), \dots, e_q(y))}. \tag{7.2}$$

Here $\mathrm{res}(x_i) = v y_i$, and

$$\bar{\sigma}_w = \mathrm{res}(\sigma_w) = v^{\ell(w)} \mathfrak{S}_w(y; 0). \tag{7.3}$$

In particular the bar notation includes the orientation power $v^{\ell(w)}$.

The Schubert classes form a basis of the entire cohomology Mackey functor, not only of its upper value. Each class σ_w defines an H-module map $\Sigma^{-\ell(w)\lambda} H \to F(X_+, H)$. Their finite wedge gives a map Θ from the free suspended module indexed by S_q. Compare Θ with the equivalence obtained by dualizing (3.4), and order both bases by degree. Its entry in row i, column j lies in $R^{(d_j - d_i)\lambda}$. By (2.6), it is zero if $d_j < d_i$, a scalar if $d_j = d_i$, and a scalar multiple of $a^{d_j - d_i}$ otherwise. Restriction kills the strictly triangular entries. The diagonal blocks identify two ordinary degreewise bases and hence are invertible. After factoring out these blocks, the remaining strictly triangular matrix is nilpotent and is inverted by a finite geometric series. Thus Θ is an equivalence of genuine H-modules. Evaluating at either orbit proves

$$\underline{H}^\alpha_G(X) = \bigoplus_{w \in S_q} \underline{R}^{\alpha - \ell(w)\lambda} \sigma_w. \tag{7.4}$$

Restriction and transfer are

$$\mathrm{res}(c\sigma_w) = \mathrm{res}_R(c) \bar{\sigma}_w, \qquad \mathrm{tr}(b \bar{\sigma}_w) = \mathrm{tr}_R(b) \sigma_w, \tag{7.5}$$

the second equality being Frobenius reciprocity. These formulas and (7.1) specify every transfer. The Weyl action at G/e is trivial: the action on X lies in a connected torus, and G preserves the representation orientations. The double-coset identity is therefore $\mathrm{res}\,\mathrm{tr} = q = 0$.

Fixed-point restriction i^* and the subgroup map res^G_e are different maps. At G/e, the map induced by $X^G \hookrightarrow X$ sends each y_i to zero and has kernel $(y_1, \dots, y_q)$. That kernel is not square zero: $y_1^2 \neq 0$ for $q \geq 3$ by the ordinary monomial basis. The square-zero claim of Theorem 1.2 concerns G/G.

7.2 An example: $q = 3$

In this subsection set $q = 3$. The dual-character roots are $(r_1, r_2, r_3) = (0, -1, 1)$ and $RO(C_3) = \mathbb{Z}\{1, \lambda\}$. Write $x = x_1$ and $y = x_2$. Eliminating $x_3 = -x - y$ from Theorem 1.1 gives

$$H^\star_G(X) = \frac{R[x, y]}{(x^3 - a^2x,\ y^2 + xy + x^2 - a^2)}. \tag{7.6}$$

Indeed, for $f_x = x^3 - a^2x$ and $f_y = y^2 + xy + x^2 - a^2$, one has $e_2 + a^2 = -f_y$ and $e_3 = f_x - xf_y$. The two successive monic relations give the basis $1, x, y, xy, x^2, x^2y$.

Our composition convention gives $s_1s_2 = 231$ and $s_2s_1 = 312$. For the Schubert basis put

$$A = \sigma_{s_1}, \quad B = \sigma_{s_2}, \quad C = \sigma_{s_1s_2}, \quad D = \sigma_{s_2s_1}, \quad F = \sigma_{w_0}.$$

The top polynomial in (5.4) specializes to $F = xy(x + a)$; applying the divided differences gives

$$A = x, \quad B = x + y + a, \quad C = xy, \quad D = x(x + a), \quad F = xy(x + a). \tag{7.7}$$

For example, reduction using (7.6) gives

$$A^2 = D - aA, \qquad B^2 = C - aB, \qquad AB = C + D.$$

The first identity follows from $D = x^2 + ax$. For the second, $B^2 + aB - C = y^2 + xy + x^2 - a^2 = 0$ in characteristic 3; and $AB = C + D$ follows directly from the definitions.

Order the fixed flags as $123, 213, 132, 231, 312, 321$ and the basis as $1, A, B, C, D, F$. Substituting $x_i|_{p_v} = r_{v_i}a$ gives

$$T = M\,\mathrm{diag}(1, a, a, a^2, a^2, a^3), \qquad M = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & -1 & 0 & 0 & 0 & 0 \\ 1 & 0 & -1 & 0 & 0 & 0 \\ 1 & -1 & 1 & -1 & 0 & 0 \\ 1 & 1 & -1 & 0 & -1 & 0 \\ 1 & 1 & 1 & -1 & -1 & 1 \end{pmatrix}. \tag{7.8}$$

This matrix is triangular over $\mathbb{F}_3$. Thus

$$\ker i^* = J_1A \oplus J_1B \oplus J_2C \oplus J_2D \oplus J_3F, \qquad \Gamma/\operatorname{im} i^* = \left(\frac{aR}{a^2R}\right)^{\oplus 2} \oplus \frac{aR}{a^3R}.$$

For $f = (f_1, \ldots, f_6)$ in this fixed-point order, computing $M^{-1}f$ over $\mathbb{F}_3$ gives

$$f \in \operatorname{im} i^* \quad \Longleftrightarrow \quad \begin{cases} f_1 - f_2 \in aR, \\ f_1 - f_3 \in aR, \\ f_1 + f_2 - f_3 - f_4 \in a^2R, \\ f_1 - f_2 + f_3 - f_5 \in a^2R, \\ -f_1 + f_2 + f_3 - f_4 - f_5 + f_6 \in a^3R. \end{cases}$$

The element $\mu_{1,1}A$ is a nonzero restriction-kernel class of integer degree 2. The tuple $f = (0, 0, 0, 0, 0, a)$ is the $q = 3$ specialization of Remark 6.1: it satisfies every edge condition but violates the last displayed condition, since its left-hand side is $a \notin a^3R$.

7.3 Ordinary and orbit-space cohomology

Forgetting the action and the orientation grading gives

$$H^*(X;\mathbb{F}_q) = \mathbb{F}_q[y_1,\dots,y_q]/(e_1(y),\dots,e_q(y)), \qquad |y_i| = 2.$$

Its Poincare polynomial is $\sum_w z^{2\ell(w)}$. The representation-graded calculation also determines $H^*(X/G;\mathbb{F}_q)$.

Corollary 7.1. *The constant-coefficient orbit-space ring satisfies*

$$H^0(X/G;\mathbb{F}_q) = \mathbb{F}_q, \qquad H^1(X/G;\mathbb{F}_q) = 0,$$

and, for $n \geq 2$,

$$\dim_{\mathbb{F}_q} H^n(X/G;\mathbb{F}_q) = \#\{w \in S_q : \ell(w) \geq \lceil n/2 \rceil\}. \tag{7.9}$$

Every product of two positive-degree classes is zero.

Proof. Integer-graded Bredon cohomology with constant coefficients is the cohomology of the orbit space [2, p. 33 and §2]. The splitting gives $H^n_G(X) = \bigoplus_w R^{n-\ell(w)\lambda}\sigma_w$. If $n = 2j \geq 2$, this group has basis $\mu_{j,\ell(w)-j+1}\sigma_w$ with $\ell(w) \geq j$. If $n = 2j+1 \geq 3$, it has basis $\nu_{j,\ell(w)-j}\sigma_w$ with $\ell(w) \geq j+1$. Equations (2.3)–(2.4) show that these are all possibilities. Degree 0 has only the unit and degree 1 vanishes. All positive-degree coefficients lie in N, so their products vanish by $N^2 = 0$. □

For $q = 3$ the Betti numbers in degrees 0 through 6 are $1, 0, 5, 3, 3, 1, 1$. These are orbit-space Betti numbers, whereas the ordinary flag variety has $1, 2, 2, 1$ in its even degrees.

A Coefficient conventions and transfer

We justify the nonzero part of (7.1) and its compatibility with the negative-cone normalization. In the subgrading, $\mathrm{res}(u) = v$ and $\mathrm{res}(a) = \mathrm{res}(\kappa) = 0$ by underlying dimension. The degrees of $\nu_{j,k}$ and $\mu_{j,k}$ have underlying dimensions $1-2k$ and $2-2k$, respectively. Only $\mu_{j,1}$ could have nonzero restriction.

For $j \geq 1$ consider

$$S(j\lambda)_+ \longrightarrow S^0 \longrightarrow S^{j\lambda}.$$

Its connecting map identifies the top degree $2j-1$ of the free sphere with $\widetilde{H_G}^{2j}(S^{j\lambda})$ at both orbits: the adjacent positive integer coefficient groups of a point vanish. The quotient of the free sphere is the oriented lens space $L^{2j-1}(q)$. On it, subgroup restriction and transfer agree with pullback and cohomological transfer for the covering

$$S^{2j-1} \longrightarrow L^{2j-1}(q). \tag{A.1}$$

The free-space identification is given in [2, Proposition 3.2]. To identify the maps, we use cellular cochains: restriction pulls back along the covering, and transfer sums its lifts. Pullback of the integral top class is multiplication by q, whereas transfer sends the sphere orientation class to the lens-space orientation class. The latter can be seen by evaluating the transferred cocycle on the base fundamental cycle: the sum of its lifts is the sphere fundamental cycle. The covering transfer construction is described in [10, §3.G]. Modulo q, pullback is zero and transfer is nonzero.

Naturality of the connecting maps gives the same statements after desuspending by $j\lambda$. The resulting upper group is $R^{2j-j\lambda} = \mathbb{F}_q\mu_{j,1}$, and the lower generator is v^{-j}. Thus $\mathrm{res}(\mu_{j,1}) = 0$ and

$\mathrm{tr}(v^{-j}) = c_j\mu_{j,1}$ for $c_j \in \mathbb{F}_q{}^{\times}$. For $j > 1$, reciprocity gives $u\,\mathrm{tr}(v^{-j}) = \mathrm{tr}(v^{1-j})$; equation (2.5) then gives $c_j = c_{j-1}$. Multiplying all $\nu_{j,k}$ and $\mu_{j,k}$ by this common scalar preserves their module relations and makes $\mathrm{tr}(v^{-j}) = \mu_{j,1}$. For $n \geq 0$, reciprocity gives $\mathrm{tr}(v^n) = u^n\,\mathrm{tr}(1) = 0$. Multiplication by the units ε_r gives (7.1) in every dimension-zero representation degree. Other underlying coefficient degrees vanish. This proves all coefficient restriction and transfer formulas used in the paper.

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